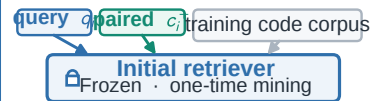


① Failure Candidate Harvesting



full-corpus ranking
high-ranked
unpaired candidates

#1	c_a	unpaired
#2	c_b	unpaired
#3	c_i^+	paired
#4	c_z	
...	c_r	

remove paired and
same-URL code

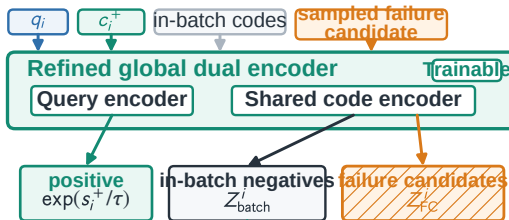
top-M pool
 $M=32$

retain top m
 $m=8$

**sample one
failure candidate**
at each update

cached once · no dynamic re-mining

② Failure-Calibrated Contrastive Refinement



$$\mathcal{L}_{\text{ref}} = \mathcal{L}_{\text{base}} + \lambda_f |B|^{-1} \sum \mathcal{L}_{\text{FC}}^i$$

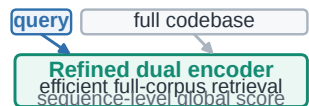
$$\mathcal{L}_{\text{FC}}^i = \log \left(1 + \sum_{c \in S_i} \exp[-\mu(q_i, c)/\tau_f] \right)$$

score margin

● — ● → **small / negative margin**
larger loss contribution

global dual-encoder retrieval is preserved

③ Dual-Granularity Reranking



top-K candidates

#1	c_a	S_g
#2	c_b	S_g
...	c_K	S_g

sequence-level
global score

token-level
MaxSim
top-k only
0 trainable params

MaxSim: latency / memory cost
**query-wise standardize
and fuse scores**

final ranking